GROUP 2 • OBJECT-ORIENTED PROGRAMMING

ABSTRACTION IN C++

Hiding Unnecessary Details

Doungue Pavel (Proposed Leader) • Banadzem Joy Berinyuy
SONE KELSON Makenzie • Toune Karl • KONCHIE TONUE PIERRE G.

1. Introduction

What does abstraction mean?

- A fundamental principle of object-oriented programming (OOP).
- Shows important features while hiding unnecessary internal details.
- Users focus on what a system does—not how it does it.

KEY IDEA: Show essential actions; hide complex implementation

2. Why Abstraction Matters

Purpose and benefits

- Reduces complexity and makes software easier to understand.
- Creates a simple interface for users and other code.
- Protects internal data from direct access.
- Makes maintenance easier when implementation changes.

3. Abstraction in Everyday Life

We use abstraction every day

ATM

Choose withdraw or check balance; bank processing stays hidden.

CAR

Use steering wheel and pedals without managing engine details.

PHONE

Tap call or message; network operations happen behind the scenes.

4. Abstraction in C++

Public interface vs private implementation

PUBLIC • INTERFACE

```
checkBalance(pin)
withdraw(pin, amount)
```

PRIVATE • HIDDEN

```
balance
authenticate(pin)
internal update logic
```

Public methods are the interface; private data and helpers remain inside the class.

5. ATM Demo & Conclusion

C++ code structure and demonstration

CORE C++ STRUCTURE

```
class ATM {
private:
    int balance;
    bool authenticate(int pin);
public:
    void checkBalance(int pin);
    void withdraw(int pin, int amount);
};
```

Private members hide details. Public methods let the user perform ATM actions.

DEMO FLOW

- Enter demo PIN: 1234
- Check balance: 50,000 FCFA
- Withdraw and see balance update
- Exit the menu

TAKEAWAY

Abstraction makes complex systems easier to use and maintain.

THANK YOU • QUESTIONS?